URBANYX

ურბანული სივრცითი ანალიზი
ყოვლისმომცველი ანგარიში

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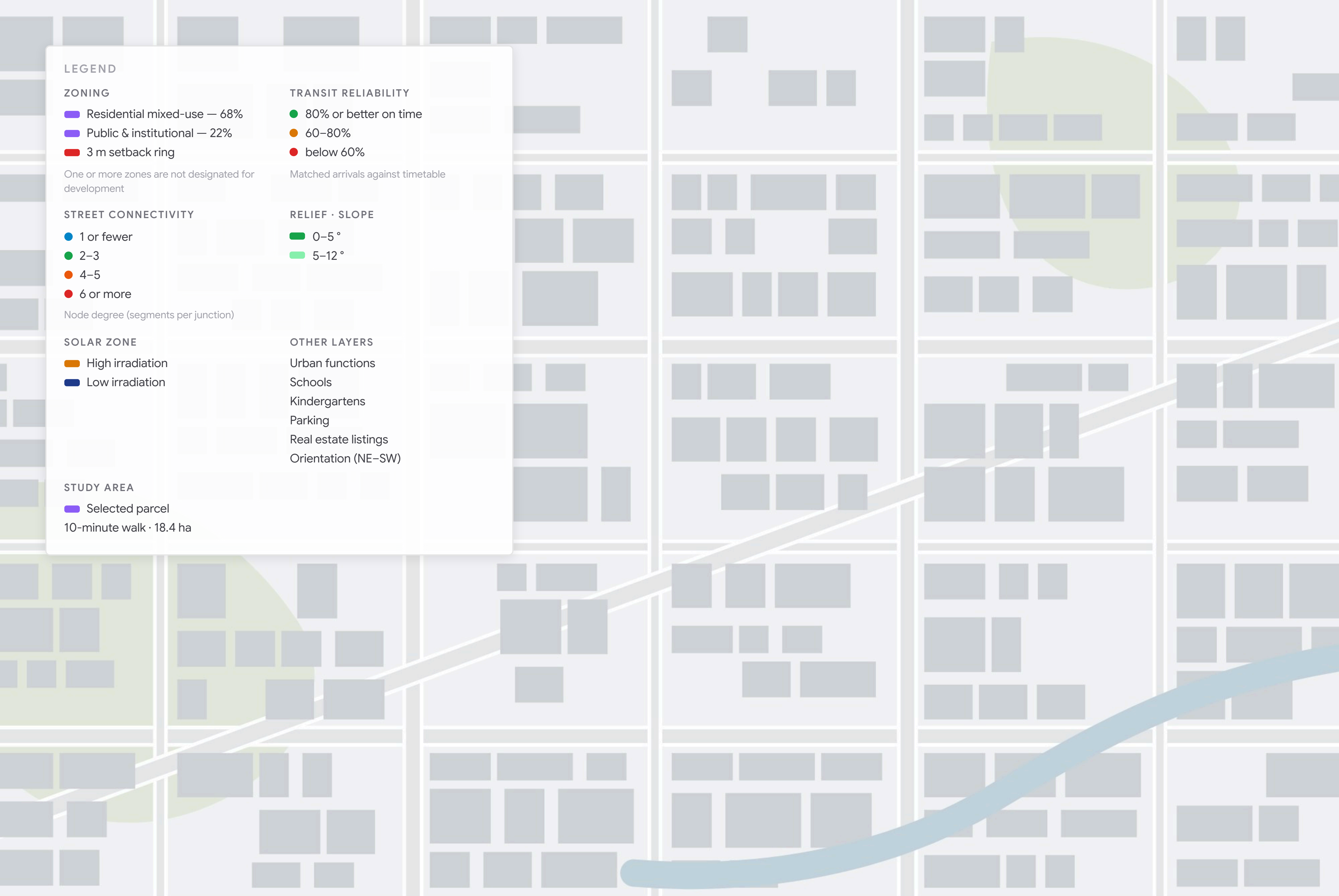
2026

Z·axis
escaping flatland



Study area and selected parcel. Basemap © Mapbox, © OpenStreetMap contributors.

Summary

At a glance, followed by the reasoning behind the figures.

74/100
LIVABILITY
Grade B

₾3,840
FLAT PRICE
median / m²

23%
TREE CANOPY
of study area

34.6°C
SURFACE TEMP
mean

2,480 m²
PARCEL AREA
01.14.09.023

This parcel scores 74 out of 100 (grade B) on the Urban Livability Index.

74
Urban Livability Index

B
good

Education	82	weight 1.0
Transit	71	weight 1.0
Parking	58	weight 0.75
Diversity	76	weight 1.0

How to read this: the index blends walkable access to amenities, public transport, land-use diversity and on-street parking into one 0–100 score (A best, F worst). Components are weighted as shown and averaged over those with data. Definitions are in Methodology.

Findings

Only the analyses active in this export are reported.

Ownership

Parcel code	01.14.09.023
Area	2,480 m²
Parcel type	Non-agricultural
Ownership type	Private
Address	Chavchavadze Avenue 37, Vake, Tbilisi
Registered	14 March 2019
Owner(s)	Kartuli Development LLC

Kartuli Development LLC · ID 405123456 · legal entity

Zoning & development limits

This parcel spans multiple zoning categories. Parameters are calculated per zone, proportional to that zone's share of the parcel area, then summed.

ZONE	SHARE	K1	K2	K3	FOOTPRINT	FLOOR AREA	GREENING	HEIGHT
Residential mixed-use	68%	0.50	2.50	0.20	843	4,215	337	5 fl
Public & institutional	22%	0.40	1.60	0.25	218	874	137	4 fl
Total (m²)					**1,061**	**5,089**	**474**	**4 fl**

3 m setback ring: 312 m² (13% of parcel) — excluded from the buildable footprint above. One or more zones carry K1 = 0; that portion is not designated for development.

Parcel compliance — Article 16

Minimum parcel area	required 500 m²	actual 2,480 m²	✓ Meets
Minimum width	required 18 m	actual 41.3 m	✓ Meets
Minimum depth	required 25 m	actual 22.8 m	✕ Below
Maximum building height	limit 5 floors	—	

Dimensions are taken from the minimum-area oriented bounding rectangle of the parcel geometry; requirements per the Tbilisi Land-Use Master Plan, Article 16.

Construction permits

Latest application	02-24-1187 (of 2 on this parcel)
Address	Chavchavadze Ave 37
Nomenclature	Multi-apartment residential building, 5 floors
Registered	11 Feb 2024
Issued	3 Jun 2024
Result	Approved with conditions

⚠ **Flag —** this permit appears to conflict with the Tbilisi 2019 Urban Masterplan: the permitted use is not among those allowed in this zone. Verify against the current masterplan before relying on it.

Street network & morphology

Connectivity	412 segments · 38.2 km · mean node degree 3.00, max 3
Orientation	1204 street ways, dominant axis NE–SW
Urban functions	Retail 31% · Food 22%

Relief · slope · aspect

Active layer	Slope
Statistics	mean 9.4° · max 34.1°

Energy potential

Solar	41% above 4 kWh/m²/day
Wind	3.4 m/s mean · 118 W/m² · 6,200 kWh/yr (5 kW ref.)

Climate & land cover

Tree canopy	23% covered
Land surface temperature	34.6°C mean

Mobility & access

88% on time
Matched arrivals, last 30 days

3 areas · ~210 cars
On-street parking capacity

Transit reliability	88% on-time, 10% late (>5 min) across 4,200 matched arrivals at 1 stop-route pairs
Parking split	2 free · 1 paid; 210 car spaces, 4 taxi, 2 loading

Nearby amenities — walkable catchment

Schools	3
Kindergartens	1
Transit stops	11 — 3 routes (4, 12, 37) · reliability B (88% on-time)
Parking areas	3 (2 free · 1 paid) — capacity ~210 cars
Land-use diversity	76/100 (Shannon SDI, normalised)

Real estate market — catchment

PROPERTY TYPE	MEDIAN PRICE / M²	LISTINGS
Flat	₾3,840	284
House	₾3,120	41

392 listings in the catchment, collected over the preceding 30 days. Asking prices, not recorded transactions.

Methodology

Only the analyses active in this export are defined here — nothing else is carried.

Urban Livability Index

A weighted mean of up to four components, each normalised to 0–1 with a saturating curve and reported on a 0–100 scale: education access (weight 1.0), public transport (1.0, moderated by a route reliability factor), land-use diversity (1.0) and on-street parking (0.75). Weights are renormalised over whichever components have data. Grades: A 80+, B 70–79, C 60–69, D 50–59, E 40–49, F below 40. Counts saturate rather than scale linearly, so a fifth school adds less than a second one.

Walkable catchment

A street-network isochrone around the site (Mapbox Isochrone API) — real walking distance, not a straight-line radius. Large parcels and drawn areas of interest are analysed over their own geometry; running the accessibility isochrone replaces the default catchment with the wider one.

Land-use diversity

Shannon diversity index over amenity-category shares in the catchment, normalised to 0–100. Higher values indicate a more even mix of functions, not more of them.

Transit reliability

The share of arrivals within −60...+300 s of schedule across the catchment's stops, weighted by observation count, over roughly 30 days of archived vehicle positions sampled every 2 minutes. Stops with fewer than 30 matched observations are excluded. Graded A 80%+ down to F below 40%.

Real estate market

Median asking prices for the selected property type across the catchment. Sale medians are per m^2; rent medians are the full monthly price. These are asking prices, not recorded transactions.

Zoning coefficients

K1 is the maximum building footprint ratio, K2 the maximum floor-area ratio and K3 the minimum greening ratio, each defined per functional zone. Where a parcel spans several zones each coefficient is applied to that zone's share of the parcel area and the results summed. Indicative height is K2 ÷ K1, rounded down. Statutory minimums for area, width and depth follow the Tbilisi Land-Use Master Plan, Article 16; width and depth are measured from the parcel's minimum-area oriented bounding rectangle, not its street frontage.

Relief

Slope and aspect are derived on the fly from a digital terrain model over the study area. Slope classes are fixed bands, not quantiles, so they are comparable between reports.

Solar potential

Irradiation modelled from the terrain's slope and aspect against the sun path for this latitude. It describes the ground and roof surface, and accounts for neither shading by adjacent buildings nor local cloud climatology.

Wind

Mean wind speed and power density at the study area from Global Wind Atlas / Open-Meteo reanalysis. Annual yield is a reference figure for a 5 kW turbine and is not a siting assessment.

Tree canopy

Share of the study area classified as tree cover in ESA WorldCover at 10 m resolution. At that resolution individual street trees are not resolved, so dense street planting reads lower than it appears on the ground.

Land surface temperature

Mean land surface temperature from Landsat 8 thermal bands at 30 m, over cloud-free summer passes. This is the temperature of the ground and roof surfaces, which runs warmer than air temperature.

Data availability

These layers rely on external services. When one is unavailable the affected component is omitted and, where applicable, the index renormalises over the components that remain; such omissions are noted in the relevant section.

Sources

Collected in order of appearance; only sources actually used are listed.

Ownership & registration	National Agency of Public Registry (NAPR), Georgia
Zoning	Tbilisi Municipality functional-zone WFS; K-coefficient limits per zone
Statutory requirements	Tbilisi Land-Use Master Plan, Article 16
Construction permits	Tbilisi Architecture Service (docs.tbilisi.gov.ge); ms.gov.ge spatial permit registry
Street network & functions	© OpenStreetMap contributors (Overpass API)
Elevation	Digital terrain model; slope and aspect derived on the fly
Solar	Slope/aspect irradiation model on the DTM
Wind	Global Wind Atlas / Open-Meteo
Tree canopy	ESA WorldCover 10 m (2021)
Land surface temperature	Landsat 8, 30 m resolution
Transit reliability	TTC vehicle positions archived every 2 min; arrivals matched to timetable (on-time −60…+300 s); stops with <30 observations excluded
Parking	Tbilisi Municipality on-street parking dataset
Education facilities	Open municipal datasets
Amenities	Public schools & kindergartens (municipal open data); transit stops & routes (Tbilisi Transport Company); on-street parking (Tbilisi Municipality); land-use POIs © OpenStreetMap via Overpass API
Real estate	Aggregated public listings, 30-day window
Basemap	© Mapbox, © OpenStreetMap contributors
Catchment	10-minute walking isochrone (Mapbox Isochrone API)